

SQL

Def **Structured Query Language (SQL)** lets you access and manipulate databases. The standard language for manipulating databases

SQL is a standard of the American National Standards Institute

Def **Query** In CS, a query is a request directed to a database or information system to retrieve specific information

Functions of SQL:

- SQL can execute queries against a database
- SQL can retrieve data from a database
- SQL can insert records in a database
- SQL can update records in a database
- SQL can delete records from a database
- SQL can create new databases
- SQL can create new tables in a database
- SQL can create stored procedures in a database
- SQL can create views in a database
- SQL can set permissions on tables, procedures, and views

DEFINITIONS

Def **Server** = Something that is meant to be communicated with over networks. Servers can communicate with another computer program and/or a database. A server is a computer or a system that provides resources, services, or data to other devices (clients) over a network.

- Servers use **database management systems (DBMS)** via a network.

Databases are Excel spreadsheets that can be tied together through keys. (An Excel spreadsheet on steroids). Databases are buckets of tables.

If using **SQL on a website** you will need a **RDBMS**

Def **RDBMS** = Relational Database Management System which stores database objects like tables.

- Ex: AWS has Amazon Relational Database Service (RDS)

Def **Tables** = A collection of related data entries (made of columns and rows)

Def **Fields** = Smaller entries within columns. Fields maintain specific information about every record (aka row) in a table

Def **Records** = Also known as a row is a individual entry that exists in a table (horizontal entity)

Def **Columns** = contains all necessary information associated with a specific field (vertical entity)

What you need for website: _____

1. A RDBMS database (MS Access, SQL Server, MySQL, Amazon RDS, Oracle Database)
2. Using some server-side scripting language (PHP or ASP)
3. Using SQL to get the data you want
4. Using HTML / CSS to style pages

Using SQLServer (Microsoft) _____

Lots of time will be concentrated under the Databases tab.

- Within your specific database, that is where you will spend 90% of your time
“New Query”

1st = Download data

2nd = Open script / data

3rd = Click create objects (Make sure you are NOT running against master)

Querying data from SQL Server _____

GENERAL SYNTAX _____

Select, update, delete, insert, where

- SELECT = extracts data from a database (selects the data from a database). Shows you how to query data against a single table
 - “Select Top 1000 rows” → Then you will write your queries
 - SELECT DISTINCT: Statement used to return only distinct (different) values
 - SELECT COUNT(DISTINCT Country) FROM Customers; // to return number of countries [For Microsoft: `SELECT Count (*) AS DistinctCountries FROM (SELECT DISTINCT Country FROM Customers);`]
 - SELECT DISTINCT = Useful for removing duplicates
- UPDATE = updates the data in a database
- DELETE = deletes the data from a database ALSO KNOWN AS DROPPING
- INSERT INTO = inserts new data into a database
- CREATE DATABASE = creates a new database
- ALTER DATABASE = modifies a database
- CREATE TABLE = creates a new table
- ALTER TABLE = modifies a table
- DROP TABLE = deletes a table
- CREATE INDEX = creates an index (search key)
- DROP INDEX = deletes an index
- WHERE = Used to extract only the records that fulfill a specified condition. It filters the rows in the output of a query based on one or more conditions
 - Example: `SELECT * FROM Customers WHERE Country='Mexico';`

- Example: `SELECT * FROM Customers WHERE CustomerID=1;`
- You can use operators (< , > , =, BETWEEN, IN, <>, etc.) with the WHERE clause
- ORDER BY = Sorts the result based on values in a specified list of columns (To sort the data)
 - Example:


```
SELECT * FROM Products
ORDER BY Price
```
- OFFSET FETCH = Limits the number of rows returned by a query
- SELECT TOP = Limits the number of rows or percentage of rows returned in a query's result set
- DISTINCT = Selects distinct values in one or more columns of a table
- Subquery = queries nested within another statement like SELECT, INSERT, UPDATE, or DELETE

EXAMPLES

Example: To select all records from a table called 'Customers' use:

```
SELECT * FROM Customers;
```

Continue here: <https://www.sqlservertutorial.net/getting-started/load-sample-database/>
https://www.w3schools.com/sql/sql_orderby.asp

Sources: W3Schools,

<https://www.youtube.com/playlist?list=PL82C6-O4XrHfZoh2ZH7-HCPyh9oHeYPnz>